Qingyang Liu

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Personal Website: https://xxxco.github.io

Education

Duke University, School of Medicine, Durham, NC — *Aug. 2024 – May 2026*
Master of Biostatistics

University of Washington, Seattle, WA — *Sept. 2020 – June 2024*
BS in Applied Computational Mathematical Science

Experience

Biostatistics Research Analyst, **Duke University Rohit Singh Lab** — *Oct. 2024 - Present*

- Built reproducible statistical pipelines in R and Python integrating 30+ biological datasets for disease gene prioritization
- Applied graph algorithms (PageRank, diffusion, embeddings) with cross-validation to assess stability of gene rankings
- Optimizing single-cell integration with GeoSketch pseudobulking and DE-based filtering for disease gene prioritization
- Building integrative models combining scRNA-seq with public network data to support discovery of therapeutic targets

Statistical Analyst, Duke Center For AIDS Research — *Oct. 2024 - Present*

- Conducted phylogenetic clustering of HIV-1 genomic regions across subtypes B to study transmission dynamics
- Applied statistical validation methods (Tree Certainty All, V-measure) to assess clustering stability and reproducibility
- Determined optimal genetic distance thresholds and demonstrated the utility of the pol region for HIV-1 monitoring
- Compared clustering outputs across regions and subtypes, providing evidence to guide public health threshold selection

Algorithm Engineering Intern, Ping'an Technology — *July 2023 – Sept. 2023*

- Processed raw video data with FFmpeg and optimized parameters to improve quality for algorithm development
- Automated analysis of 80,000+ video files in Python, extracting metadata for database management and quality checks
- Built and organized databases containing 11,000+ minutes of processed data to facilitate AI training and evaluation

Projects

Pilot Trial Designing Software, Duke University — *Aug. 2024 - Present*

- Designed a statistical framework to evaluate feasibility of pilot studies for future randomized clinical trials
- Applied Bayesian methods to estimate dropout probabilities, generate credible intervals, and assess study uncertainty
- Established composite decision rules combining acceptability and precision criteria to determine trial continuation
- Built an interactive Shiny app to calculate Bayesian assurance and support sample size planning for trial design

Multimodal Single-Cell Integration, University of Washington — *Mar. 2023 – June 2023*

- Built predictive models (Linear Regression, LGBM, MLP) to examine RNA–protein co-variation in single-cell dataset
- Trained models on 70,000+ CD34+ hematopoietic stem and progenitor cell samples collected from multiple donors
- Applied Python libraries (SCANPY, SHAP) to interpret outputs, quantify feature contributions, ensuring reproducibility
- Conducted single-cell RNA-seq analysis and generated visualizations to identify and communicate biological patterns

Vaccine Reservation System, University of Washington — *Dec. 2021*

- Developed SQL-based pipelines for hospitals and suppliers to update vaccine availability and distribution
- Simulated COVID-19 vaccine allocation strategies and supply chain management across pharmacies and hospitals
- Implemented user-level modules for appointment scheduling, access monitoring, and vaccine availability tracking
- Developed user-facing modules supporting secure login, appointment scheduling, and availability monitoring

Skills

Statistical Programming: R (tidyverse, Shiny), Python, SQL, Java
Statistical & Analytical Methods: A/B Testing, Bayesian inference**,** Regression Modeling, Exploratory data analysis
Healthcare & Bioinformatics Tools: Scanpy, SHAP, SEACells, scLink, Decipher, RaxML, HIV-Trace, TreeCluter
Research Tools: Git, Docker, LaTeX, FFmpeg